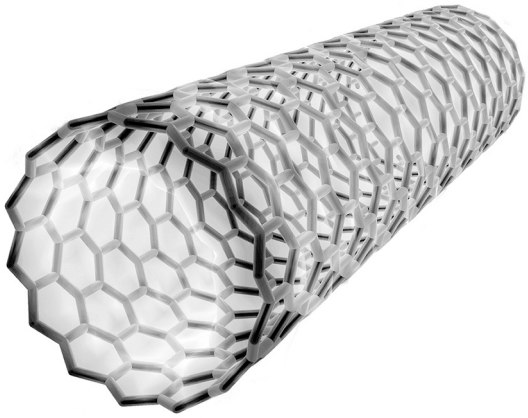
Do not write outside the box

0 8 Carbon nanotubes are cylindrical fullerenes.

Figure 11 represents the structure of a carbon nanotube.

Figure 11



0 8 . 1 Describe the arrangement of carbon atoms in the nanotube shown in **Figure 11**.

[1 mark]

_____

_____

0 8 . 2 Nanotubes are used in electronics.

Give **one** other use of nanotubes.

[1 mark]

_____

_____

Do not write outside the box

0 8 . 3 A nanotube contains 2380 carbon atoms.

Calculate the number of moles of carbon in this nanotube.

The Avogadro constant is 6.02×10^{23} per mole.

[2 marks]

Number of moles of carbon = _____ mols

0 8 . 4 Explain why carbon nanotubes can conduct electricity.

Refer to bonding between carbon atoms in your answer.

[3 marks]

7

END OF QUESTIONS